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# **(PROPOSED): Use of Snowflake Native AI Capabilities**

**ID:**

**Decision Type:** Strategic / Architectural

**Decision Owner:** Data Integration and Architecture

**Decision Level:** Platform / Enterprise

**Status:** Proposed

**Review Cadence:** TBD

## **Executive Summary**

Establishes how Snowflake's native AI capabilities(Cortex.AI) will be incorporated into the enterprise data platform in a controlled, low-risk manner. The decision positions AI as an assistive intelligence layer—used to accelerate data understanding, profiling, and metadata generation—while preserving human oversight, auditability, and deterministic data processing. Autonomous AI-driven transformations and decisioning are explicitly deferred until data maturity, confidence thresholds, and governance requirements are met. This approach enables near-term value from AI while protecting trust, regulatory posture, and long-term architectural integrity.

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## **1. Decision Statement**

Determine how and where Snowflake native AI capabilities (e.g., Cortex, Document AI, Copilot-style interfaces) may be used within the enterprise data platform to improve insight and efficiency while maintaining governance, auditability, and acceptable risk.

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## **2. Decision Context**

Snowflake has introduced native AI capabilities that embed probabilistic inference directly within the data warehouse environment. These capabilities enable advanced reasoning over governed data without external data movement but introduce non-deterministic behavior into traditionally deterministic data pipelines.

The organization operates in a risk-sensitive environment and prioritizes:

- Trustworthy and explainable data outputs
- Clear separation between analysis and execution
- Human accountability for high-impact decisions

The decision must balance near-term analytical benefit against long-term operational and regulatory risk.

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### 3. Decision

Snowflake native AI capabilities will be used **only to support human decisions**, not to make or execute decisions autonomously.

Specifically:

- AI may generate **insights, annotations, classifications, and draft artifacts**.
  - AI outputs are **inputs into human decision-making**, not decision outcomes.
  - Any AI output that influences persistent data or downstream logic requires explicit validation and promotion.
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### 4. Decision Scope

#### In Scope (Permitted Uses)

- Data and schema profiling
- Metadata inference and enrichment
- Classification and summarization
- Draft generation of analytical or design artifacts (e.g., STM proposals)
- Read-only analysis and non-persistent inference

#### Out of Scope (Excluded Uses)

- Autonomous data transformations
  - AI-driven business rule execution
  - Direct modification of production datasets
  - End-user natural language decision interfaces without semantic governance
  - Black-box or non-auditable decision logic
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## 5. Decision Logic (Why This Decision Was Chosen)

This decision reflects the following logic:

- **Value is highest** when AI reduces human cognitive load in understanding complex data.
- **Risk is lowest** when AI operates in advisory, non-persistent roles.
- **Cost of failure is acceptable** when outputs are observable, explainable, and reversible.
- **Cost of failure is unacceptable** when AI acts autonomously on regulated or high-impact data.

Therefore, AI is constrained to decision support roles where humans retain accountability.

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## 6. Decision Inputs

The decision is informed by:

- Platform capabilities and limitations of Snowflake native AI
  - Regulatory and governance requirements
  - Current data maturity and schema stability
  - Known characteristics of probabilistic inference systems
  - Organizational readiness for human-in-the-loop workflows
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## 7. Decision Outputs

The decision produces:

- Approved usage patterns for Snowflake AI capabilities
  - Explicit constraints on automation and persistence
  - Governance expectations for AI-assisted workflows
  - A baseline against which future AI expansion can be evaluated
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## 8. Decision Consequences

### Positive Consequences

- Faster insight into data quality and structure
- Reduced manual effort in early-stage analysis

- Clear accountability and audit trails
- Lower operational and regulatory risk

## **Negative Consequences / Tradeoffs**

- Slower adoption of autonomous AI features
  - Continued reliance on human validation
  - Some AI capabilities deferred or underutilized
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## **9. Decision Metrics (How We Know This Is Working)**

- Reduction in analyst time spent on profiling and metadata tasks
  - Adoption of AI-assisted insights in design and review workflows
  - Zero unauthorized AI-driven data modifications
  - No audit findings attributable to AI-assisted processes
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## **10. Review & Reconsideration Criteria**

This decision should be revisited if:

- Data schemas reach sustained stability
- AI confidence, explainability, and rollback mechanisms mature
- Regulatory guidance changes
- Snowflake materially changes AI execution or governance controls

Any expansion into autonomous or automated decisioning requires a new Decision Record.

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## **Decision Principle (Summary)**

**AI informs decisions; humans make decisions; systems execute decisions.**

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